SECTION A

Answer **all** questions.

1. A hydrocarbon is either cyclopentane or pent-1-ene.

Use the **Data Booklet** to choose a characteristic infrared absorption that will positively identify the compound as pent-1-ene. [1]

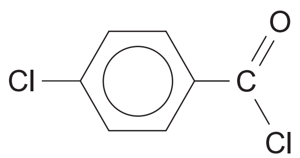
....

....

2. Give the **name** of a compound of formula C_4H_8 that will decolourise acidified potassium manganate(VII). [1]

....

3. Complete the equation for the reaction shown below. [1]

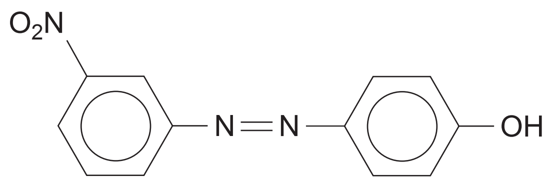


$+ \ H_2O \longrightarrow \qquad\qquad +$

4. Give the structure of a compound of molecular formula $C_4H_8Br_2$ whose low resolution ^{1}H NMR spectrum consists of three peaks. [1]

Examiner only

5. The formula of an azo dye, produced by reacting a diazonium compound with phenol, is shown below.



Write the formula of the amine that is used to produce this diazonium compound. [1]

6. Ethanal reacts with hydrogen cyanide to give 2-hydroxypropanenitrile, $CH_3CH(OH)(CN)$.

(a) State the type of mechanism occurring during this reaction. [1]

....

(b) Draw the structures of the two enantiomers of 2-hydroxypropanenitrile. [1]

1410U401 03

Examiner only

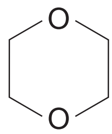
7. Three isomers have the molecular formula $C_4H_8O_2$.

(a) (i) Give the structure of an isomer that gives bubbles of carbon dioxide when tested with aqueous sodium hydrogencarbonate. [1]

(ii) Give the structure of an isomer that is both a ketone and a secondary alcohol. [1]

(b) Describe and explain the high resolution ^1H NMR spectrum of 1,4-dioxane.

The position of the signal(s) is **not** required. [1]



1,4-dioxane

....

....

10